✅ 1. Check if the Array is Sorted in Ascending Order

Problem Statement:

Write a program to check whether the given array is sorted in strictly ascending order or not. An array is considered sorted in ascending order if each element is smaller than or equal to the next element.

Input:

n space-separated integers (array elements)

Output:

- Print "Array is sorted in ascending order" if the array is sorted
- Otherwise, print "Array is not sorted"

Example:

Input:

1 2 3 4 5

Output:

Array is sorted in ascending order

Input:

3 5 2 9 10

Output:

Array is not sorted

✅ 2. Separate 0s and 1s in an Array

Problem Statement:

Given an array consisting only of 0s and 1s, write a program to rearrange the array such that all the 0s appear before all the 1s. The order of elements within the groups doesn't matter.

Input:

n space-separated elements (only 0s and 1s)

Output:

- A single line with the rearranged array where all 0s come before 1s

Example:

Input:

1 0 1 0 1 0

Output:

0 0 0 1 1 1

Explanation:

The input array has 3 zeros and 3 ones. Rearranged version brings all zeros to the beginning and ones to the end.

✅ 3. Find the Smallest Positive Number in an Array

Problem Statement:

Given an array of integers, write a program to find the **smallest positive number** (greater than zero). Ignore zeros and negative numbers.

Input:

- n space-separated integers

Output:

- Print the smallest positive number in the array
- If no positive number exists, print "No positive number found"

Example:

Input:

-2 0 5 -3 8 1 -7

Output:

1

Explanation:

The positive numbers are 5, 8, and 1. Among them, 1 is the smallest.

✅ 4. Count Elements Within a Given Range

Problem Statement:

Write a program to count how many elements in the array lie between two given numbers a and b (inclusive). You may assume that a ≤ b.

Input:

- n space-separated integers (array elements)
- Two integers a and b (the range boundaries)

Output:

- Print the count of elements that lie in the range [a, b] inclusive

Example:

Input:

4 7 2 10 15 3

Boundary/range => 4 10

Output:

4

Explanation:

The numbers within the range 4 to 10 are: 4, 7, 10, and 3 → Only 4 elements fall in this range.